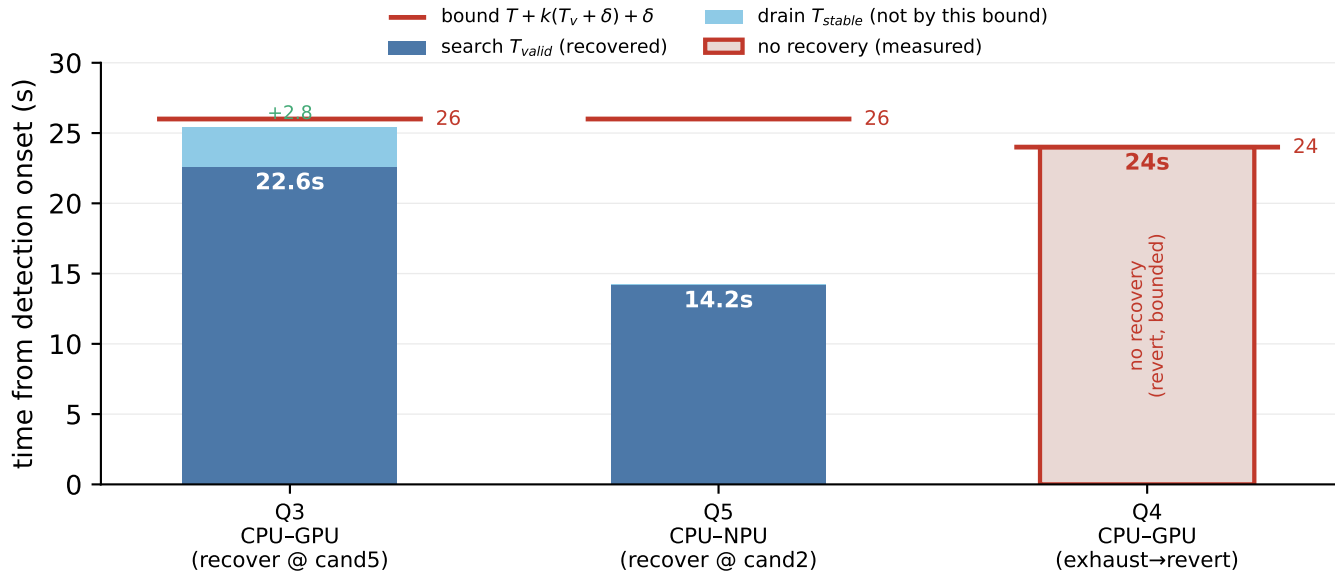




detection onset = first violation, except Q5 (stable already violates → controller enabled at burst) · windows: Q3/Q5 ~107s, Q4 100s